# Lab 5: Design a bio-instrumentation amplifier for EOG

November 24, 2023

| Name | Student # | Lab Contribution |
|---|---|---|
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**Question 1: Plot of amplitude vs angle of gaze theta [4]**

To get a plot of amplitude vs. angle of gaze, we had our subject look at sticky notes 5° apart from -30° to +30° over a span of 40 seconds. We aimed to have the subject move from one sticky note to the next every three seconds. The resulting EOG signal is shown in the amplitude vs time plot below:

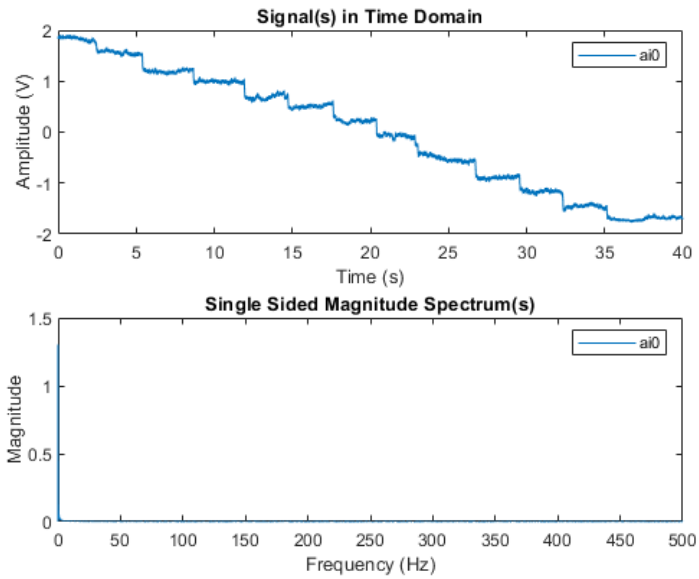


*Figure 1. Amplitude vs. time*

As observed from this graph, there is an observable step at intervals where the gaze is switched. We "binned" the data using the sub-intervals marked below, and averaged all of the bins:

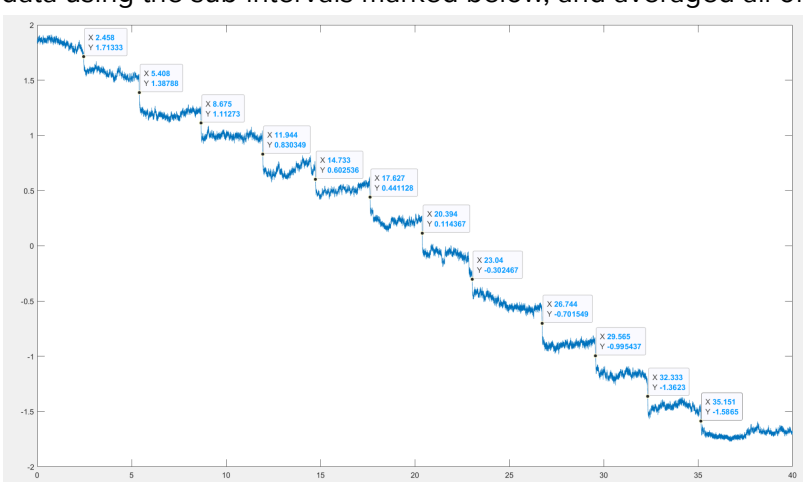


*Figure 2. Bins used to segment eye movements for varying theta*

The following code was used:

```
gazearray =
[mean(data(1:2458)),mean(data(2458:5408)),mean(data(5408:8675)),mean(data(8675:
11944)),mean(data(11944:14733)),mean(data(14733:17627)),mean(data(17627:20394))
,mean(data(20394:23040)),mean(data(23040:26744)),mean(data(26744:29565)),mean(d
ata(29565:32333)),mean(data(32333:35151)),mean(data(35151:40000))]
```

The following amplitude versus the angle plot was then developed:

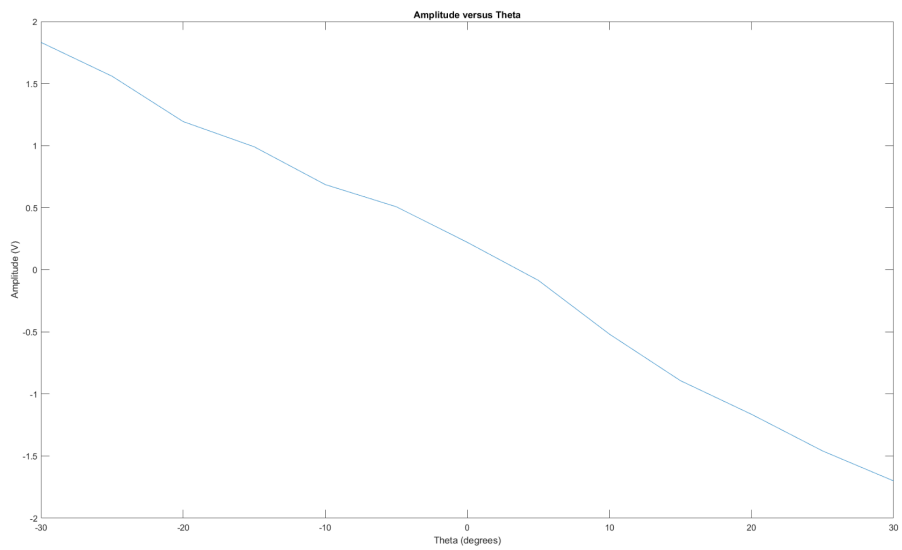


*Figure 3. Amplitude vs theta*

Using a linear regression model in MATLAB, the relationship between amplitude and theta found was y = -0.064807x + 0.043008, and the $R^2$ value found was 0.989. The linear equation represents the modelled relationship between the horizontal eye angle and recorded voltage. If the eye were to continuously rotate from -30° to 30° rather than holding at 5° intervals, an output similar to the fit would be produced. $R^2$ represents the accuracy of the linear regression model, with a value of 1 indicating a perfectly accurate model and a value of 0 indicating a complete dissociation between the two variables. Since the $R^2$ for this model is 0.989, it indicates that the model very accurately predicts the relationship between theta and amplitude.

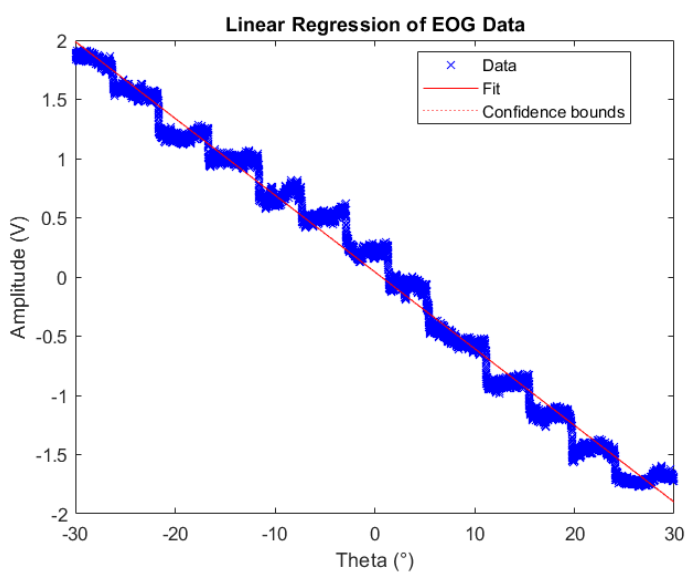


*Figure 4. Plot of Linear Regression Model*

**Question 2: Discuss differences between the theoretical and actual EOG signal/data [1]**

Theoretically, a nearly linear relationship can be observed between the horizontal angle of gaze and EOG signal. This arises from the dipole structure of the eyes - where a positive charge can be found at the cornea and negative at the retina. [1] When the eyes move, the orientation of the dipole changes which creates an electrical potential that the EOG can measure. When assessing horizontal eye movements, as the subject looks left or right, the positive charge of the cornea moves in that respective direction creating a detectable potential difference. This relationship between the horizontal angle and EOG signal appears linear because the magnitude of the electrical potential is proportional to the degree of eye movement - therefore, as the eyes move 30° left or right, the EOG signal increases in amplitude.

However, in actuality, a number of factors create noise and distortions that prevent the development of a linear relationship. Firstly, the effect of the recording equipment is equivalent to approximately a 1° shift in eye movement. [1] This is compounded by a lack of accuracy from the EOG when detecting very small (eye movements of less than 1° - 2°) or large movements (such as a sudden sweep that can be greater than 30°) as they are difficult to record and produce amplitudes that don't proportionally reflect eye position. In addition to this, our subject wasn't following an object continuously across their gaze; they were jumping from sticky note to sticky note at discrete time intervals which would lead to a step output on the graph, and could possibly taint results depending on the speed at which the eye jumps between targets.

Given the calculated $R^2$ value of 0.989, it can be determined that the actual EOG data very closely reflects the aforementioned characteristics of theoretical EOG data.

**Question 3: Plot the vertical and horizontal EOG signals with the blink artifacts. Describe the blink artifacts and compare them between the horizontal and vertical EOG signals. [3]**

The blink artifacts can be seen on the plots below (Figure 5 and Figure 6) as a rapid downward deflection followed by a slow return to the baseline (eyes closing during the blink and then reopening). This manifests as a distinctive waveform with a consistent pattern, as the blinking occurred in constant increments. The magnitude of the horizontal EOG's are less than the magnitude of the vertical EOG's. This is due to the fact that the horizontal EOG measures movement side to side which may not generate a significant impact on the electrical potential compared to the vertical movements. Blink artifacts with a vertical EOG involve a more pronounced vertical component with the opening and closing of eyelids.

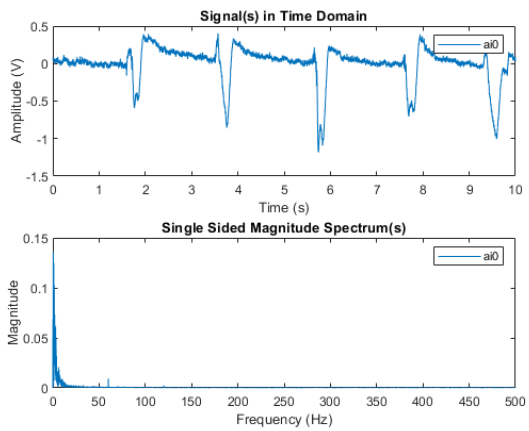


*Figure 5. Horizontal EOG Signal (blink artifacts)*

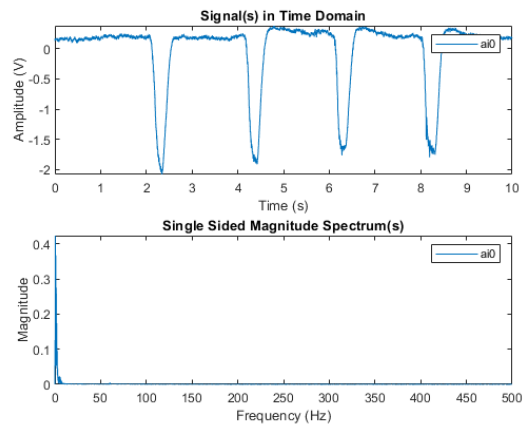


*Figure 6. Vertical EOG Signal (blink artifacts)*

**Question 4: TA Demonstration of mouse control with EOG:**

Two videos have been attached showing successful control of the mouse with vertical and horizontal EOG signals, along with blink detection to control mouse clicks instead of left mouse click. The task demonstrated to the TA was to close tabs using blink detection. These can be found in the dropbox where we submitted this document.

## References

[1] A. G. Webb, *Principles of Biomedical Instrumentation*. Cambridge, UK: Cambridge University Press, 2019.

## Appendix

### Lab Step 1: Building and Testing the Circuit

*Table 1: Gain vs. Frequency (Hz) for signals between DC and 100Hz used to test the circuit using the wave generator as input, omitting the 2nd amplification stage and using a 10 mV amplitude input signal.*

| Frequency (Hz) | Vin (mV) | Vout | G=Vout/Vin |
|---|---|---|---|
| 1 | 10 mV | 560 mV | 56 |
| 5 | 10 mV | 560 mV | 56 |
| 10 | 10 mV | 560 mV | 56 |
| 20 | 10 mV | 500 mV | 50 |
| 25 | 10mV | 440 mV | 44 |
| 30 | 10 mV | 420 mV | 42 |
| 45 | 10 mV | 360 mV | 36 |
| 60 | 10 mV | 320 mV | 32 |
| 80 | 10 mV | 260 mV | 26 |
| 100 | 10 mV | 240 mV | 24 |

### Code

Segmenting and plotting amplitude vs theta:

```
gazearray =
[mean(data(1:2458)),mean(data(2458:5408)),mean(data(5408:8675)),mean(data(8675:
11944)),mean(data(11944:14733)),mean(data(14733:17627)),mean(data(17627:20394))
,mean(data(20394:23040)),mean(data(23040:26744)),mean(data(26744:29565)),mean(d
ata(29565:32333)),mean(data(32333:35151)),mean(data(35151:40000))]
xaxis=linspace(-30,30,13)
plot(xaxis,gazearray)
xlabel('Theta (degrees)')
ylabel('Amplitude (V)')
title('Amplitude versus Theta')
```

linearfit.m:

```
y = data(1:39*Fs+1);  % remove final second of data
angle = 60/39 * time(1:39*Fs+1) - 30;  % convert time to eye angle
mdl = fitlm(angle, y);  % create linear regression model
figure,
plot(mdl),
title("Linear Regression of EOG Data"),
xlabel("Theta (°)"); ylabel("Amplitude (V)");
```